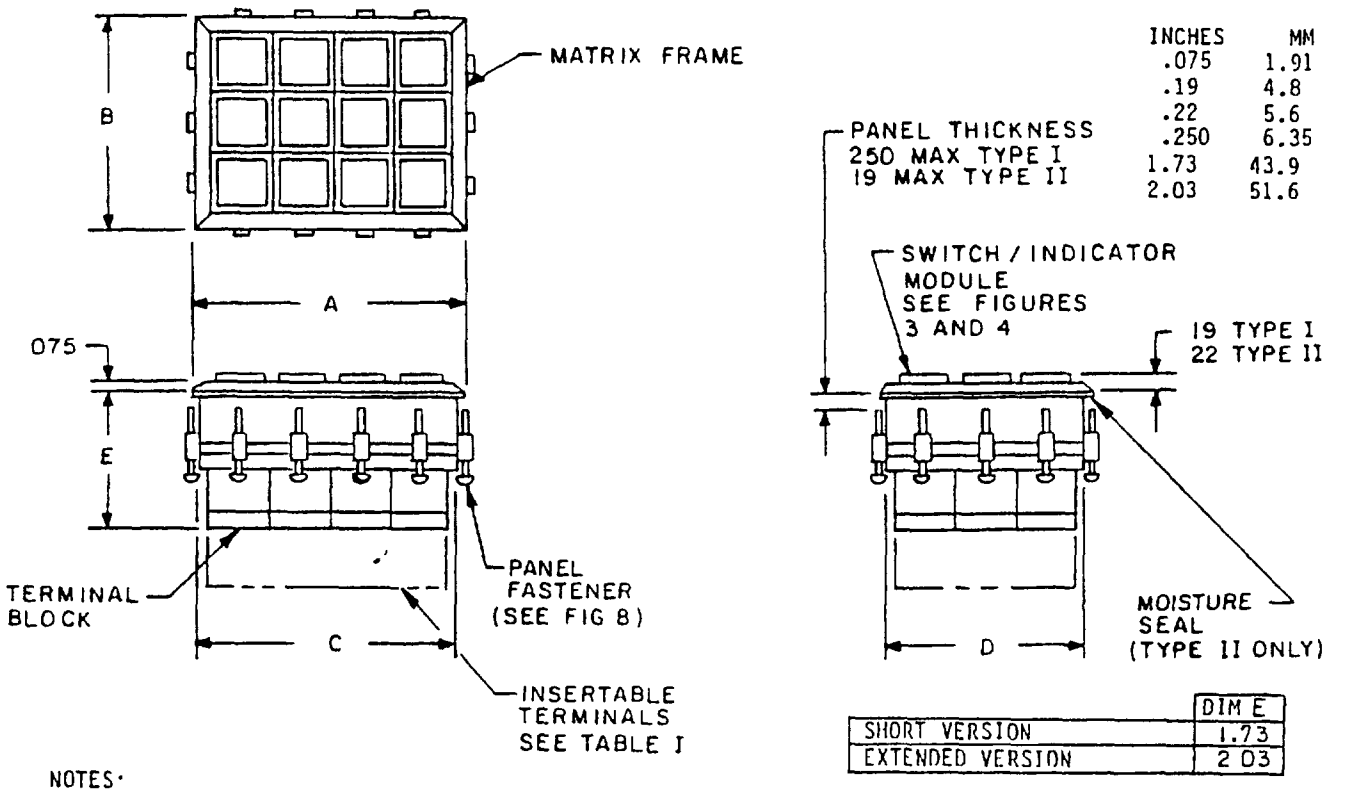
MILITARY SPECIFICATION SHEET

SWITCH, MULTISTATION, PUSHBUTTON, (ILLUMINATED MATRIX ARRANGEMENT) PLUG-IN MODULE, EMI/RFI SHIELDED, DRIPPROOF, STYLE SM11

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-S-24317.



INCHES	MM
.075	1.91
.19	4.8
.22	5.6
.250	6.35
1.73	43.9
2.03	51.6

	DIM E
SHORT VERSION	1.73
EXTENDED VERSION	2.03

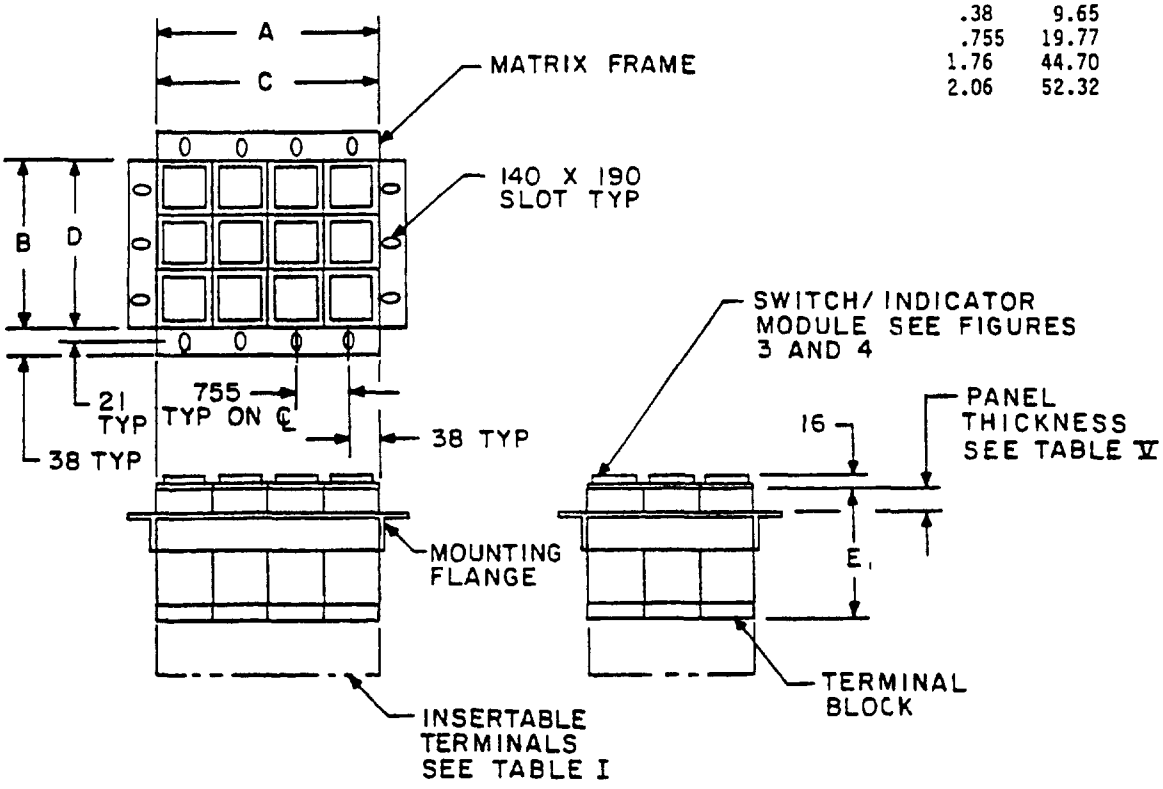
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.015 (0.38 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Design configuration is optional within the envelope dimensions shown.
4. Recommended panel cutout. See table IX.
5. For dimensions A, B, C, and D, see table IX.

FIGURE 1. Matrix frame assembly, unsealed - type I, sealed - type II (with or without EMI/RFI shielding.

(B) denotes changes

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INCHES	MM
.140	3.56
.16	4.06
.190	4.83
.21	5.33
.38	9.65
.755	19.77
1.76	44.70
2.06	52.32



	DIM E
SHORT VERSION	1.76
EXTENDED VERSION	2.06

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.015 (0.38 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Design configuration is optional within the envelope dimensions shown.
4. Recommended panel cutout. See table IX.
5. For dimensions A, B, C, and D, see table IX.

FIGURE 2. Matrix frame assembly, rear mount, unsealed - type III (with or without EMI/RFI shielding).

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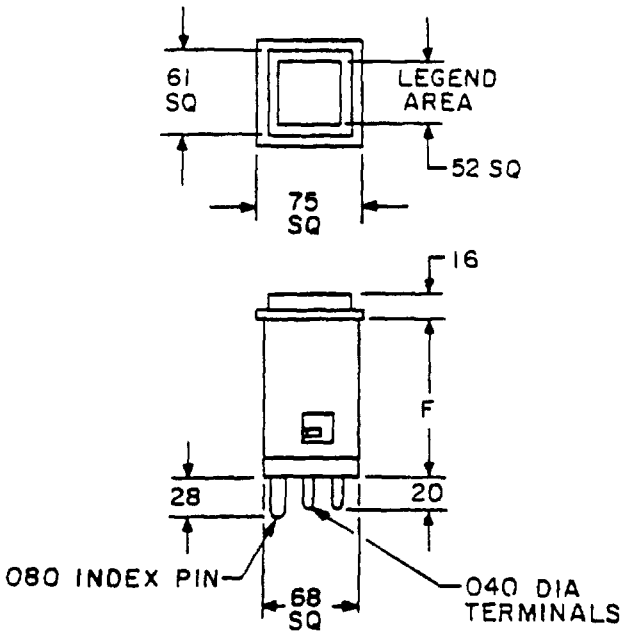


FIGURE 3. Plug-in switch/indicator module per MIL-S-22885/101 (with or without EMI/RFI shielding).

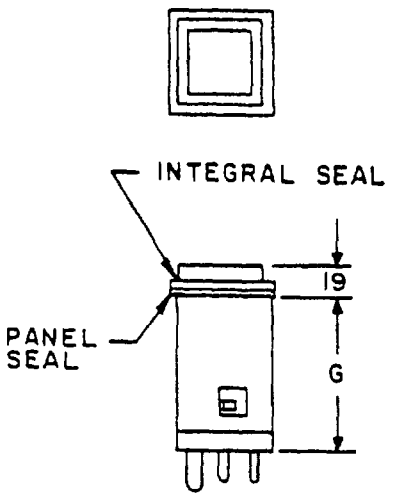


FIGURE 4. Panel dripproof sealed plug-in switch/indicator module per MIL-S-22885/101 (with or without EMI/RFI shielding)

(B)

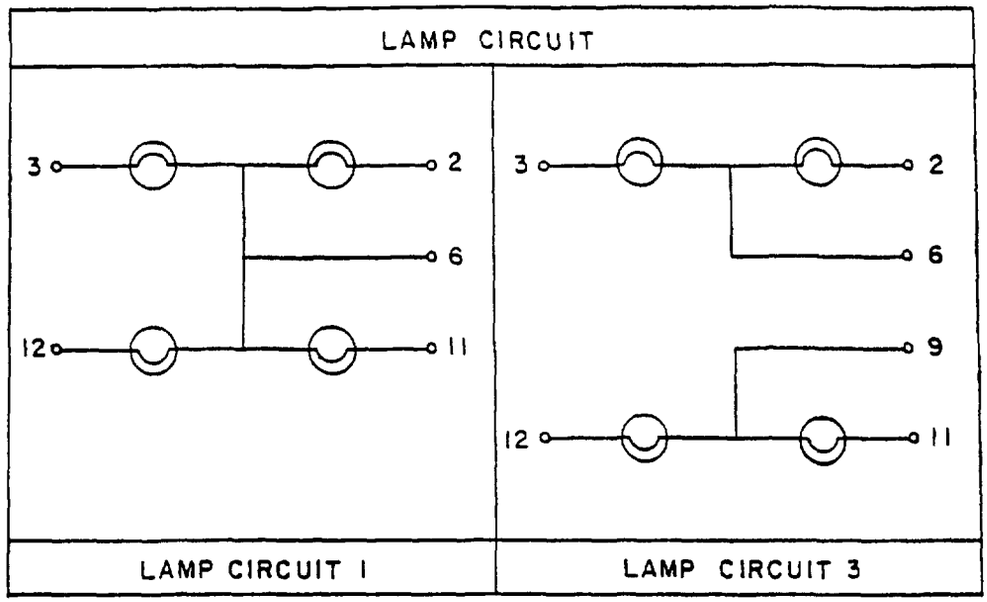
	DIM F	DIM G
SHORT VERSION	1.20	1.17
EXTENDED VERSION	1.50	1.47

INCHES	MM
.040	1.02
.080	2.03
.16	4.1
.19	4.8
.20	5.1
.28	7.1
.52	13.2
.61	15.5
.68	17.3
.75	19.1
1.17	29.7
1.20	30.5
1.47	37.3
1.50	38.1

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Design configuration is optional within the envelope dimensions shown.

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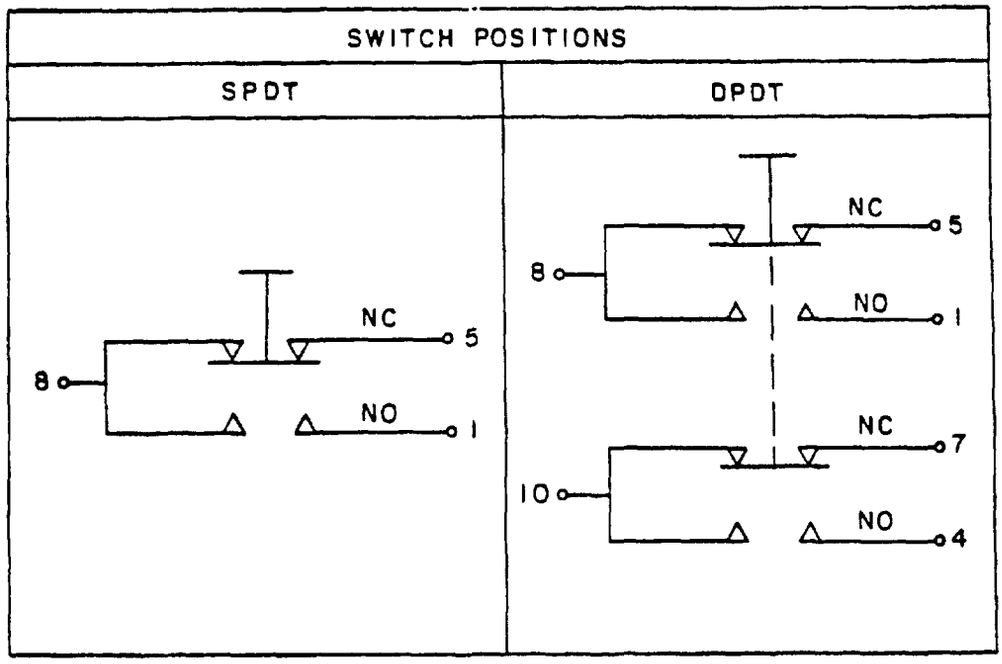
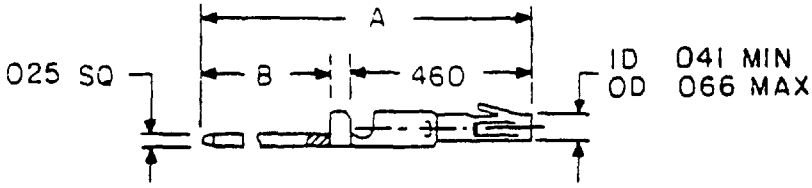
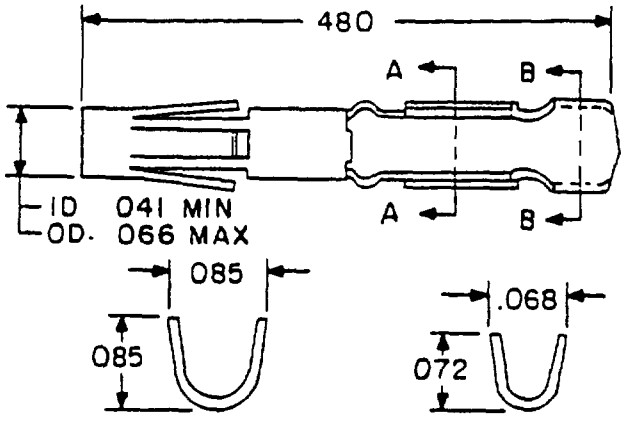


FIGURE 5. Lamp and switch circuit identification.

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WIRE WRAP TYPE (STYLES A, B, AND C)



Inches	mm
.025	0.64
.041	1.04
.066	1.68
068	1.73
.072	1.83
.085	2.16
.257	6 53
460	11.68
471	11.96
.480	12.19
.685	17.40
.767	19.48
981	24.92
1.195	30.35

CRIMP TYPE (STYLE D)

TABLE 1

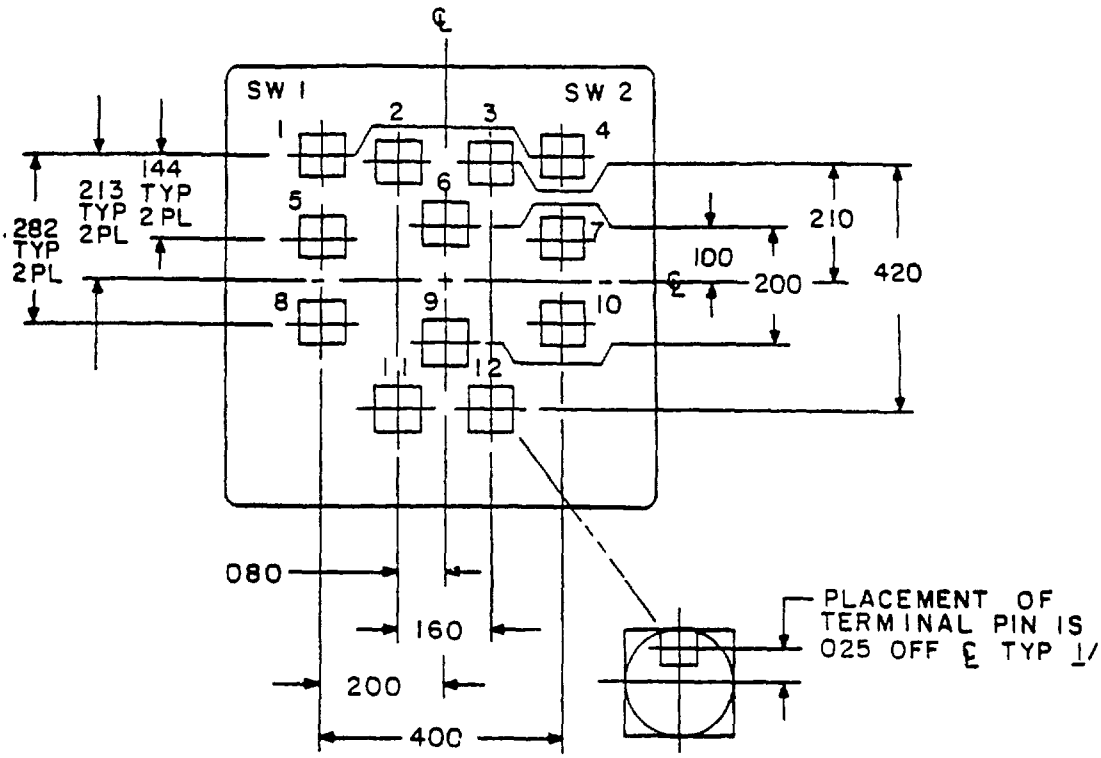
Terminal style	Terminal type	Maximum termination per post	Dimensions	
			A	B
A 1/	Wire wrap	1	.767	.257
B	Wire wrap	2	981	.471
C	Wire wrap	3	1.195	685
D	Crimp	2/	---	---

1/ Style A terminal may be used for printed circuit board mounting. See figure 7.
2/ Style D terminal accepts 20-24 AWG wire.

NOTES.
1 Dimensions are in inches. Metric equivalents are given for information only.
2. Unless otherwise specified, tolerances are ± 010 (0 25 mm) for three place decimals and ± 03 (0.8 mm) for two place decimals.
3. Use with extraction tool MSC products (CAGE 96182) part number 581-960 or equivalent.
4. Crimp with MSC products (CAGE 96182) hand tool part number 581-959 or equivalent

(8) FIGURE 6 Insertable terminals

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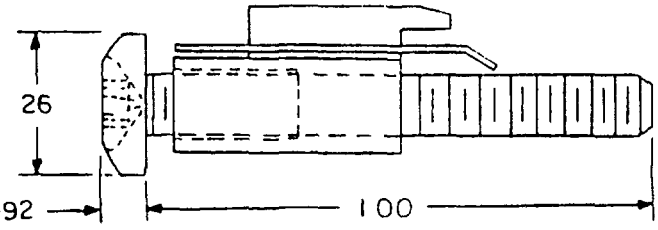
INCHES	MM
.025	0.64
.080	2.03
.100	2.54
.144	3.66
.160	4.06
.200	5.08
.210	5.33
.213	5.41
.282	7.16
.400	10.16
.420	10.67

NOTES:
1/ Terminal may be installed in any four positions: upper, lower, right, or left.
2. Dimensions are in inches.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

FIGURE 7. Switch terminal identification.

(REAR VIEW - TERMINAL BLOCK)

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INCHES	MM
.092	2.34
.26	6.6
1.00	25.4

NOTES
1. Screw is MS51957-34.
2 For use with type I and type II only
3 Design optional within envelope dimensions shown

FIGURE 8. Panel fastener.

REQUIREMENTS

Dimensions and configurations. See figures and tables.

Complete multistation switch assembly shall consist of.

1. Multistation matrix frame (figures 1 and 2 and tables IV and V).

2. Plug-in switch indicator module in accordance with MIL-S-22885/101, when specified in contract or purchase order.

(B) 3. Insertable terminals, when specified in contract or purchase order (see figure 6 and table VI).

(B) 4. Four lamps per station (T-1 submidget flange base incandescent), when specified in contract or purchase order. 1/

(B) Acquisition of government spares Matrix frame, plug-in switch indicator module, terminals and lamps shall be acquired separately. Provisioning documentation shall show parts breakdown by associated military part number.

Matrix frame:

Type I. Includes terminal block and fasteners.

Type II: Includes terminal block, fasteners and panel seal.

Type III: Includes terminal block and flanges.

Plug-in switch/indicator module Includes switch (except indicator modules), lamp housing, and lens assembly (in accordance with MIL-S-22885/101).

Frame material: Aluminum alloy, QQ-A-200/9.

Frame finish

Chemical film/black paint - Type I per MIL-C-5541.
Type II per MIL-C-5541.

Chemical film - Type III per MIL-C-5541.

Electrical· See tables II and III.

Switch actuation force 2 to 5 pounds.

1/ Industry Lamp No. 6180 is recommended (5 volt, .060 amps, .15 MSCP). Other lamps are available with various power requirements and illumination results. When using higher wattage lamps, additional heat sinking and air flow must be provided (use extended version).

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Simultaneity: Not applicable.

Color and luminance: In accordance with MIL-S-22885/101.

Solderability: Applicable (style A only).

Lamp retention: Applicable.

Torque strength of plunger: Not applicable.

Shock: Method I, in accordance with MIL-S-22885.

Seal: Dripproof, when specified.

RFI: Test in accordance with MIL-S-22885/101, when specified.

Thermal shock: During high temperature portion of thermal shock, all four lamps shall be energized with full rated voltage. Total lamp wattage per station not to exceed 1.2 watts for short version. Use extended version for lamp power requirements greater than 1.2 watts.

Terminal insertion strength: Each terminal to be tested shall be inserted and removed 5 times. After this, the force to dislodge the terminals from the terminal block must not be less than 5 pounds. The force applied shall be approximately 1 pound per second.

Marking: In accordance with MIL-STD-130.

Matrix frame part number:

Short version: M24317/11- (dash number from table IV or V).

Extended version. M24317/11E (dash number from table IV or V).

(B) Terminals and panel fastener part numbers: M24317/11- (dash number from table VI).

TABLE II. Electrical ratings - silver contacts.

Load	Sea level, 28 V dc NO or NC	70,000 feet, 28 V dc NO or NC
	(Amperes, maximum)	(Amperes, maximum)
Resistive	5.0	5.0
Inductive	3.0	2.0
Lamp	1.0	---

TABLE III. Electrical ratings - gold plated contacts.

Load	Sea level, 28 V dc	70,000 feet, 28 V dc
	(Amperes, max.)	(Amperes, max).
Resistive	1	1
Inductive	0.5	0.5

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TABLE IV. Type I and type II matrix mounted frame part number M24317/11- (dash numbers from table IV).

Dash number 1/		Matrix arrangement 2/	Number of stations	Number of fasteners per side	
Type II sealed	Type I unsealed	H V		Horiz. side	Vert. side
-201	-101	1 X 1	1	1	1
-202	-102	1 X 2	2	1	2
-203	-103	1 X 3	3	1	3
-204	-104	1 X 4	4	1	4
-205	-105	1 X 5	5	1	5
-206	-106	1 X 6	6	1	6
-207	-107	1 X 7	7	1	7
-208	-108	1 X 8	8	1	8
-209	-109	1 X 9	9	1	9
-210	-110	1 X 10	10	1	10
-211	-111	2 X 1	2	2	1
-212	-112	2 X 2	4	2	2
-213	-113	2 X 3	6	2	3
-214	-114	2 X 4	8	2	4
-215	-115	2 X 5	10	2	5
-216	-116	2 X 6	12	2	6
-217	-117	2 X 7	14	2	7
-218	-118	2 X 8	16	2	8
-219	-119	2 X 9	18	2	9
-220	-120	2 X 10	20	2	10
-221	-121	3 X 1	3	3	1
-222	-122	3 X 2	6	3	2
-223	-123	3 X 3	9	3	3
-224	-124	3 X 4	12	3	4
-225	-125	3 X 5	15	3	5
-226	-126	3 X 6	18	3	6
-227	-127	3 X 7	21	3	7
-228	-128	3 X 8	24	3	8

See footnotes at end of table.

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TABLE IV. Type I and type II matrix mounted frame part number M24317/11- (dash numbers from table IV) - Continued.

Dash number 1/		Matrix arrangement 2/		Number of stations	Number of fasteners per side	
Type II sealed	Type I unsealed	H	V		Horiz. side	Vert. side
-229	-129	3	9	27	3	9
-230	-130	3	10	30	3	10
-231	-131	4	1	4	4	1
-232	-132	4	2	8	4	2
-233	-133	4	3	12	4	3
-234	-134	4	4	16	4	4
-235	-135	4	5	20	4	5
-236	-136	4	6	24	4	6
-237	-137	4	7	28	4	7
-238	-138	4	8	32	4	8
-239	-139	4	9	36	4	9
-240	-140	4	10	40	4	10
-241	-141	5	1	5	5	1
-242	-142	5	2	10	5	2
-243	-143	5	3	15	5	3
-244	-144	5	4	20	5	4
-245	-145	5	5	25	5	5
-246	-146	5	6	30	5	6
-247	-147	5	7	35	5	7
-248	-148	5	8	40	5	8
-249	-149	5	9	45	5	9
-250	-150	5	10	50	5	10
-251	-151	6	1	6	6	1
-252	-152	6	2	12	6	2
-253	-153	6	3	18	6	3
-254	-154	6	4	24	6	4
-255	-155	6	5	30	6	5
-256	-156	7	1	7	7	1

See footnotes at end of table.

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TABLE IV. Type I and type II matrix mounted frame part number M24317/11- (dash numbers from table IV) - Continued.

Dash number 1/		Matrix arrangement 2/	Number of stations	Number of fasteners per side	
Type II sealed	Type I unsealed	H V		Horiz. side	Vert. side
-257	-157	7 X 2	14	7	2
-258	-158	7 X 3	21	7	3
-259	-159	7 X 4	28	7	4
-260	-160	7 X 5	35	7	5
-261	-161	8 X 1	8	8	1
-262	-162	8 X 2	16	8	2
-263	-163	8 X 3	24	8	3
-264	-164	8 X 4	32	8	4
-265	-165	8 X 5	40	8	5
-266	-166	9 X 1	9	9	1
-267	-167	9 X 2	18	9	2
-268	-168	9 X 3	27	9	3
-269	-169	9 X 4	36	9	4
-270	-170	9 X 5	45	9	5
-271	-171	10 X 1	10	10	1
-272	-172	10 X 2	20	10	2
-273	-173	10 X 3	30	10	3
-274	-174	10 X 4	40	10	4
-275	-175	10 X 5	50	10	5

1/ Dash numbers shown are for matrix frame with fasteners only, does not include plug-in switch/indicator module or terminals.

2/ "H" indicates number of horizontal stations; "V" indicates number of vertical stations.

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TABLE V. Type III matrix mounted frame part number M24317/11- (dash numbers from table VI).

Dash number 1/ Panel thickness				Matrix arrangement 3/	Number of stations	Number of flanges 2/
.190	.125	.090	.063	H V		
-601	-501	-401	-301	1 X 1	1	2V, 2H
-602	-502	-402	-302	1 X 2	2	"
-603	-503	-403	-303	1 X 3	3	"
-604	-504	-404	-304	1 X 4	4	"
-605	-505	-405	-305	1 X 5	5	"
-606	-506	-406	-306	1 X 6	6	"
-607	-507	-407	-307	1 X 7	7	"
-608	-508	-408	-308	1 X 8	8	"
-609	-509	-409	-309	1 X 9	9	"
-610	-510	-410	-310	1 X 10	10	"
-611	-511	-411	-311	2 X 1	2	"
-612	-512	-412	-312	2 X 2	4	"
-613	-513	-413	-313	2 X 3	6	"
-614	-514	-414	-314	2 X 4	8	"
-615	-515	-415	-315	2 X 5	10	"
-616	-516	-416	-316	2 X 6	12	"
-617	-517	-417	-317	2 X 7	14	"
-618	-518	-418	-318	2 X 8	16	"
-619	-519	-419	-319	2 X 9	18	"
-620	-520	-420	-320	2 X 10	20	"
-621	-521	-421	-321	3 X 1	3	"
-622	-522	-422	-322	3 X 2	6	"
-623	-523	-423	-323	3 X 3	9	"
-624	-524	-424	-324	3 X 4	12	"
-625	-525	-425	-325	3 X 5	15	"
-626	-526	-426	-326	3 X 6	18	"
-627	-527	-427	-327	3 X 7	21	"

See footnotes at end of table.

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TABLE V. Type III matrix mounted frame part number M24317/11- (dash numbers from table V) - Continued

Dash number 1/ Panel thickness				Matrix arrangement 3/	Number of stations	Number of flanges 2/
.190	.125	.090	.063	H V		
-628	-528	-428	-328	3 X 8	24	2V, 2H
-629	-529	-429	-329	3 X 9	27	"
-630	-530	-430	-330	3 X 10	30	"
-631	-531	-431	-331	4 X 1	4	"
-632	-532	-432	-332	4 X 2	8	"
-633	-533	-433	-333	4 X 3	12	"
-634	-534	-434	-334	4 X 4	16	"
-635	-535	-435	-335	4 X 5	20	"
-636	-536	-436	-336	4 X 6	24	"
-637	-537	-437	-337	4 X 7	28	"
-638	-538	-438	-338	4 X 8	32	"
-639	-539	-439	-339	4 X 9	36	"
-640	-540	-440	-340	4 X 10	40	"
-641	-541	-441	-341	5 X 1	5	"
-642	-542	-442	-342	5 X 2	10	"
-643	-543	-443	-343	5 X 3	15	"
-644	-544	-444	-344	5 X 4	20	"
-645	-545	-445	-345	5 X 5	25	"
-646	-546	-446	-346	5 X 6	30	"
-647	-547	-447	-347	5 X 7	35	"
-648	-548	-448	-348	5 X 8	40	"
-649	-549	-449	-349	5 X 9	45	"
-650	-550	-450	-350	5 X 10	50	"
-651	-551	-451	-351	6 X 1	6	"
-652	-552	-452	-352	6 X 2	12	"
-653	-553	-453	-353	6 X 3	18	"
-654	-554	-454	-354	6 X 4	24	"
-655	-555	-455	-355	6 X 5	30	"

See footnotes at end of table.

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TABLE V. Type III matrix mounted frame part number M24317/11- (dash numbers from table V) - Continued.

Dash number 1/ Panel thickness				Matrix arrangement 3/	Number of stations	Number of flanges 2/
.190	.125	.090	.063	H V		
-656	-556	-456	-356	7 X 1	7	2V, 2H
-657	-557	-457	-357	7 X 2	14	"
-658	-558	-458	-358	7 X 3	21	"
-659	-559	-459	-359	7 X 4	28	"
-660	-560	-460	-360	7 X 5	35	"
-661	-561	-461	-361	8 X 1	8	"
-662	-562	-462	-362	8 X 2	16	"
-663	-563	-463	-363	8 X 3	24	"
-664	-564	-464	-364	8 X 4	32	"
-665	-565	-465	-365	8 X 5	40	"
-666	-566	-466	-366	9 X 1	9	"
-667	-567	-467	-367	9 X 2	18	"
-668	-568	-468	-368	9 X 3	27	"
-669	-569	-469	-369	9 X 4	36	"
-670	-570	-470	-370	9 X 5	45	"
-671	-571	-471	-371	10 X 1	10	"
-672	-572	-472	-372	10 X 2	20	"
-673	-573	-473	-373	10 X 3	30	"
-674	-574	-474	-374	10 X 4	40	"
-675	-575	-475	-375	10 X 5	50	"

1/ Dash numbers shown are for matrix frame with flanges only and does not include plug-in switch/indicator module or terminals.

2/ 2V = two vertical flanges; 2H = two horizontal flanges.

3/ "H" indicates number of horizontal stations; "V" indicates number of vertical stations.

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(B) TABLE VI. Terminals and panel fastener dash numbers.

Dash number	Description
901	Terminal style A (see fig 6). Package of 25 terminals.
902	Terminal style B (see fig 6). Package of 25 terminals.
903	Terminal style C (see fig 6). Package of 25 terminals.
904	Panel fastener (see fig 8)
905	Terminal style D (see fig 6). Package of 25 terminals.

QUALIFICATION:

All applicants for qualification approval shall demonstrate that each of their items conform to all the requirements in the applicable documents, singularly and in combination with all the other previously qualified items, regardless of manufacturer.

Extent of qualification (group submission):

The multistation switch/indicator assembly to be used for qualification inspection shall be the 12 station (4x3) matrix. Qualification of the 12 station units shall qualify all multistation units up to and including the 50 station unit. The 12 station units for qualification shall contain the following items:

Part number		Description	Quantity
M24317/11-233	and M24317/11E233	4x3 matrix frame - type II	1 each
M22885/101-161W1W	and M22885/101E161W1W	2PDT momentary switch module	2 each
M22885/101-167W1W	and M22885/101E167W1W	2PDT alternate switch module	2 each
M22885/101-155W1W	and M22885/101E155W1W	Indicator module	8 each
M24317/11-533	and M24317/11E533	4x3 matrix frame - type III	1 each
M22885/101-119W1W	and M22885/101E119W1W	2PDT momentary switch module	2 each
M22885/101-125W1W	and M22885/101E125W1W	2PDT alternate switch module	2 each
M22885/101-113W1W	and M22885/101E113W1W	Indicator module	8 each

Group submission:

The following applies when the 12 station units (4x3 matrix), type II and type III, short and extended versions, are submitted for qualification at the same time.

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TABLE VII. Qualification inspection - group submission.

Assembly	Qualification inspection table V of MIL-S-24317		Extent of approval
	Group	Number of samples	
Schedule I Type II Short version	I II III V	3 2 (from group I) 2/ 1 (from group I) 1 (from group II) 1/	Types I and II Short version
Schedule II Type III Short version	I II	2 2 (from group I)	Type III Short version
Schedule I Type II Extended version	I II III V	3 2 (from group I) 2/ 1 (from group I) 1 (from group II) 1/	Types I and II Extended version
Schedule II Type III Extended version	I II	2 2 (from group I)	Type III Extended version

1/ One sample unit is required, and shall be subject to groups I and II and group V, seal test only.
2/ One sample unit shall be tested for RFI shielding.

Group C inspection:

Group C inspection shall be in accordance with MIL-S-24317, except that inspection shall be as shown in table VII herein.

Group A inspection: Shall be in accordance with table VIII.

TABLE VIII. Group A inspection.

Examination or test	Requirement and inspection reference
Visual and mechanical examination	MIL-S-24317
Switch actuation and circuit configuration	MIL-S-24317

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TABLE IX. Dimensions (figures 1 and 2). 1/

No. of stations (vertically or horizontally)	Matrix dimensions A and B ±.017(.43)	Cutout dimensions C and D +.030(.76) -.000
Types I and II (see figure 1)		
1	1.153 (29.29)	.973 (24.71)
2	1.908 (48.46)	1.728 (43.89)
3	2.663 (67.64)	2.483 (63.07)
4	3.418 (86.82)	3.238 (82.25)
5	4.173 (105.99)	3.993 (101.42)
6	4.928 (125.17)	4.748 (120.60)
7	5.683 (144.35)	5.503 (139.78)
8	6.438 (163.53)	6.258 (158.95)
9	7.193 (182.70)	7.013 (178.13)
10	7.948 (201.88)	7.768 (197.31)
Type III (see figure 2)		
1	.745 (18.92)	.765 (19.43)
2	1.500 (38.10)	1.520 (38.61)
3	2.255 (57.28)	2.275 (57.79)
4	3.010 (76.45)	3.030 (76.96)
5	3.765 (95.63)	3.785 (96.14)
6	4.520 (114.81)	4.540 (115.32)
7	5.275 (133.99)	5.295 (134.49)
8	6.030 (153.16)	6.050 (153.67)
9	6.785 (172.34)	6.805 (172.85)
10	7.540 (191.52)	7.560 (192.02)

1/ Dimensions are in inches. Millimeter equivalents are in parentheses and are given for general information only.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99

User activity:
Air Force - 19

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5930-F652)

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